Bijoy Joseph

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Experience:

Robert Bosch Engineering and Business Solutions	**May 2019 – present**
Robosoft Technologies	**October 2018 – January 2019**
Telyport Technologies	**June 2018 – September 2018**
Typecast Technologies (operations closed)	**November 2017 – May 2018**
TalentBridge Technologies	**November 2016 – November 2017**

Technical Skills:

Programming Languages: Java, JDK
Frameworks: Spring Boot, Spring MVC, JDBCTemplate, JOOQ, Apache Kafka
Databases: MySQL, PostgreSQL, Redis, MongoDB
Tools & Technologies: RESTful APIs, Microservices, Docker, Kubernetes, RabbitMQ
Load Balancers & API Gateways: NGINX, Zuul, Spring Cloud Gateway
Version Control: Git, GitHub, Gerrit
Cloud Platforms: AWS (EC2, RDS, S3), Azure, GCP
Testing Tools: JUnit, Mockito, Postman, JMeter
Methodologies: Agile, Scrum, TDD, CI/CD

Projects:

FACT - Remote System Image monitoring

- Designed REST APIs to read OS images and report missing sectors, used JSON and XML for data interchange.
- Integrated and modified IBM RTC APIs to specifically suit the need of the project. Wrote services to handle multiple concurrent connection requests to the tune of few thousands.
- Designed login and signup strategies to keep user session active on multiple devices simultaneously.
- Implemented file compression strategies using 7zip, created APIs to transfer files and across multiple individual modules of the project.
- Implemented mailing system and APIs to send and receive custom emails containing the status report of the file analysis – related to the project.
- Used MQTT for messaging layer, to communicate between different microservices, used Apache Kafka for stream processing (reading OS image data and sending it to a different service for analysis)
- Optimized API response times through caching mechanisms (Redis), resulting in a 50% decrease in latency. Utilized Docker containers for service deployment, ensuring consistent environments across development and production.

Technologies Used: Java8/Java11, Spring Boot, Spring Security, JDBCTemplate, PostgresSQL, Redis

File Sync Application/Remote File Storage

- Developed microservices to upload and sync file to a remote server using Java
- Integrated Dropbox and Google cloud storage APIs to get access to third party file hosting services.
- Used AMQP and a combination of MQTT for push notifications and broadcasting various status messages.

- Implemented strategies to hash the uploaded files and encrypt certain file types based on SHA256 and bcrypt hashing algorithms
- Used elastic search for file search and document search feature.
- Used MongoDB for handling unstructured data, created API gateways for handling multiple requests.
- Used NginX for reverse proxy, handled system authorization using root signed certificates, devised various strategies for cron-job tasks.
- Used various Agile methodologies, including SCRUM, Test-Driven Development (TDD), and Pair Programming, contributing to streamlined and efficient development workflows.
- Designed and implemented RESTful APIs for legacy applications, enhancing modularity and independence, thereby ensuring seamless integration with modern technologies.
- Ensured compliance with security standards (e.g., OAuth 2.0, JWT) for user authentication and data protection. Developed a comprehensive logging system using SLF4J and Logstash for error tracking and performance monitoring.
- Implemented data validation and error handling strategies to improve API reliability. Participated in Agile ceremonies, contributing to sprint planning and retrospectives to enhance team performance. Maintained API documentation using Swagger for improved developer experience.
- Integrated RabbitMQ for asynchronous processing of transaction notifications, improving system responsiveness and reliability. Conducted performance optimization using SQL query tuning and indexing strategies, resulting in a 40% increase in data retrieval speeds.

Technologies Used: Java11, MongoDB, MySQL, AMQP, NginX, OAuth, Spring Security

HMI Analytics application

- Implemented ETL pipelines to gather multiple vehicle telemetry data from different sensor types and from different vehicle vendors, processing millions of data per day enabling proper parsing and storage of data.
- Designed and implemented multiple microservices to collect different types of vehicle and hardware info, enabling applications to derive thousands of interfaces daily.
- Integrated CI/CD pipelines, Revamped legacy systems using Java and Spring Boot, improving system performance, streamlined codebase with agile development, boosting productivity, managed and mentored a team of junior developers, increasing overall proficiency.
- Designed and implemented a generic library for user management helping both internal and external projects quickly integrate mechanisms like authentication, authorisation, user registration, login, roles and permissions cutting the overall development time by 50% and improving developer productivity.
- Implemented automated report generation system which gave insights about system status and load on different services which are currently being used by the system.
- Integrated Spring Security for role-based access control, enhancing application security. Conducted performance testing using JMeter, identifying bottlenecks and optimizing query performance. Collaborated with front-end teams to ensure seamless integration of APIs with Angular and React applications.

Technologies Used: Java11, Spring Boot, Spring Security, Spring Cloud Gateway, RabbitMQ, Swagger

Education

M.Sc. (Software Technology), 2016
College: AIMIT, St Aloysius College, Mangalore, KA, India (Mangalore University)

BCA (Bachelor of Computer Application), 2012
College: Mangalore University, Mangalore, KA, India